

Group 1: Swing State Election Analysis

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[Website](#)

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Motivation: Recent elections have resulted in close races determined by narrow margins in key battleground states.

Goal: Create models to accurately predict which counties candidates should focus on in key battleground states, based on factors like population and likelihood for large voting shifts.

Key Research Questions/Hypotheses:

Q1: Which counties illustrate a combination of **electoral elasticity** and **vote volume**, maximizing the marginal utility of Democratic resource allocation?

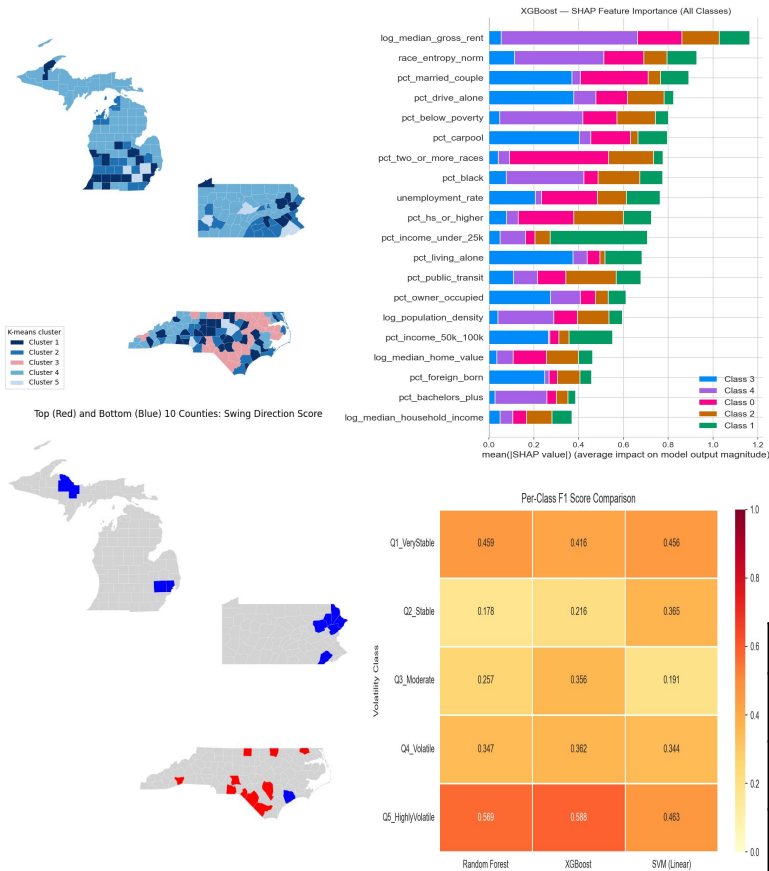
Q2: What specific demographic and economic features distinguish **"High-Volatility"** county clusters from **"Stable"** clusters?

Q3: To what extent do demographic predictors of electoral volatility generalize across state lines?

H1: **Electoral volatility** is inversely correlated with **wealth**.

H2: **Racial heterogeneity** is a stronger predictor of Democratic vote share than individual demographics.

H3: High-volatility counties will shift in the same direction as the statewide or national trend but with significantly greater magnitude, making them the highest-leverage targets for 2028.



Data: County demographic data (from NHGIS and ACS APIs) and MIT election returns. All three sources were limited to MI, NC, and PA counties for 2016, 2020, and 2024.

Models: Random Forest, XGBoost, SVM, Logistic Regression, Multiple Linear Regression, K-Means

Findings:

Recommendations:

Model	Accuracy	F1-macro	ROC-AUC	Best At
XGBoost	0.388	0.387	0.783	Best overall
SVM (Linear)	0.368	0.364	0.765	Middle classes
Random Forest	0.368	0.362	0.803	Best ROC-AUC